TECHNICAL ANALYSIS USING R

Assignment 5 – Development Stage

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Course: CA_BDA400

Assignment: A05

Repository: https://github.com/srbrar1-hub/TechnicalAnalysis/tree/main/Assignment_5

Report scope

This report describes nine base-R technical-analysis functions, how to run the submitted scripts, and the observed sample outputs. The scripts were run in RStudio, as shown in the supplied execution screenshots. No external R package is required for these implementations.

1. Implementation summary

Simple Moving Average (sma.R) — Computes the arithmetic mean over successive fixed-size windows; returns one value per complete window.

Exponential Moving Average (ema.R) — Uses multiplier 2/(period + 1), initializing with the first observation and recursively updating the weighted average.

Moving Average Convergence Divergence (macd.R) — Subtracts a long-period EMA from a short-period EMA; computes an EMA signal line and their difference (histogram).

Standard Deviation (stdev.R) — Computes the population standard deviation using squared deviations from the arithmetic mean.

Linear Regression (linreg.R) — Calculates a fitted straight line with slope, intercept and predicted values from a selected data subset.

Relative Strength Index (rsi.R) — Separates price gains and losses, calculates initial averages and applies Wilder smoothing to obtain a 0–100 oscillator.

Stochastic RSI (stoch_rsi.R) — Normalizes RSI over a rolling RSI window to generate a K line, then smooths it to obtain a D line.

Crossover (crossover.R) — Returns TRUE where the first series moves from at/below the second series to above it.

Crossunder (crossunder.R) — Returns TRUE where the first series moves from at/above the second series to below it.

2. Execution instructions

1. Download all nine .R files from the Assignment_5 GitHub folder into the same local folder. 2. Open RStudio and set the working directory to that folder (Session → Set Working Directory → Choose Directory). 3. Run each script using Source, or type source("sma.R"), source("ema.R"), and so on in the R console. 4. The MACD script depends on ema.R; the Stochastic RSI script depends on rsi.R and sma.R. Keep the dependent scripts together. 5. Inspect the printed test results in the Console.

3. Observed testing

The supplied RStudio screenshots show successful execution of all nine sample scripts without visible errors. Representative observed results: SMA begins 12.33333, 15.66667; EMA begins 10, 11, 13; MACD prints macd_line, signal_line and histogram; population standard deviation = 4.946629; regression slope = 0.7428571 and intercept = 19.06667; RSI prints initial NA entries followed by 60, 74.4 and later values; Stochastic RSI prints k_line and d_line; Crossover signals TRUE at position 4; Crossunder signals TRUE at position 6.

4. Notes and limitations

The sample tests demonstrate that the scripts execute and return plausible values; they are not exhaustive tests of every edge case. SMA returns only complete windows, while other indicators preserve their own output lengths. The standard deviation uses the population denominator n, rather than sample denominator n−1. The Stochastic RSI uses rolling normalization. The regression implementation and indicator conventions should be compared with any instructor-provided reference tests before grading.

5. Conclusion

Nine base-R technical-analysis functions were implemented, executed on sample data and uploaded to the Assignment_5 repository. They cover trend indicators, momentum indicators, volatility, regression and crossover signals. The repository contains the R source files for review and further testing.